

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Performed laboratory work in accordance with client protocols and SOPs for inhalation studies, ensuring adherence to quality standards and regulatory awareness.
- Prepared study protocols, project status reports, and final study reports, detailing experimental designs, data interpretations, and technical findings for client review.
- Organized and supervised laboratory resources for group project activities, coordinating schedules and updating project status to ensure timely delivery and efficient resource allocation.
- Reviewed and interpreted experimental data for technical quality and compliance with protocols, methods, and SOPs, identifying deviations and initiating root cause analysis for method investigations.
- Responded to client inquiries, coordinated communication, and documented meeting minutes, maintaining strong client partnerships and ensuring clear understanding of project requirements and progress.
- Assisted senior group leaders in managing experienced professionals and subordinate staff, fostering a collaborative environment and ensuring independent execution of assignments.
- Containerized laboratory simulation tools using **Docker** and deployed them to an **AWS EC2** instance for consistent testing environments, improving deployment reliability by **20%**.

SKILLS

Laboratory Operations – Inhalation studies, Aerosol characterization, Dose delivery, Protocol development, Study execution, Client communication, SOP adherence, QA metrics.

Scientific Documentation – Study protocols, Project status reports, Final study reports, Technical documentation, Method investigations, Root cause analysis, Technical writing.

Project Management – Resource organization, Scheduling, Supervision, Project status updates, Client coordination, Time management, Project solution driven.

Technical Skills – Laboratory equipment operation, Data analysis, Method optimization, AWS EC2, Docker, Python scripting, Linux shell scripting.

Leadership & Mentoring – People leadership, Mentoring junior staff, Performance management, Team supervision, Client partnership, Cross-functional collaboration.

PROJECTS

Inhalation Study Protocol & Data Analysis Platform

- Developed a Python-based system to automate the generation of standard study protocols and data summary reports for inhalation studies, reducing manual effort by approximately **40%**.
- Implemented data validation checks using **pandas** and **NumPy** to ensure accuracy and compliance of experimental results against client-defined parameters and SOPs.
- Designed a REST API using **FastAPI** to serve protocol templates and analysis summaries, facilitating easier access for project managers and scientific staff.

Laboratory Automation & Monitoring System

- Built a **Java Spring Boot** application to monitor key parameters of inhalation exposure systems, integrating with sensors and triggering alerts via **AWS Lambda**.
- Containerized the monitoring application using **Docker**, enabling consistent deployment across development and production environments on **AWS EC2**.
- Wrote comprehensive unit tests using **JUnit**, achieving over **90%** code coverage for critical monitoring and alerting functionalities.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)